

Mohamed Taha Abo Heiba

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GitHub: github.com/MoTahaAboHeiba | Portfolio: my-portfolio.mohamed-aboheiba.workers.dev

PROFESSIONAL SUMMARY

Data Engineer who builds tested, incrementally-loaded pipelines and dimensional models (star, galaxy, SCD2) that hold up under real data quality checks, across Databricks, dbt, Airflow, SQL Server, and AWS. **Huawei HCIA-Big Data certified and AWS Certified Cloud Practitioner, ranked best-performing trainee in a 30-person cohort**

PROFESSIONAL EXPERIENCE

- **AWS Cloud Data Engineer Trainee | National Telecommunication Institute (NTI)** *Oct 2025 – Dec 2025*
 - Architected multi-tier **AWS** app (EC2/RDS/VPC/S3) using **Well-Architected Framework** for fault-tolerant, HA design
 - Ensured high availability via **RDS Multi-AZ** configuration; reduced cold storage costs by implementing S3 lifecycle policies to Glacier
- **Big Data Engineer Trainee | National Telecommunication Institute (NTI)** *Jul 2025 – Sep 2025*
 - **Ranked best-performing trainee** in a 30-person cohort; earned exam voucher and passed Huawei HCIA-Big Data certification exam
 - Built Spark batch pipelines on Hadoop and Kafka, Spark streaming across 15 labs, enabling scalable batch and real-time processing
- **Data Engineer Trainee | Digital Egypt Pioneers Initiative (DEPI)** *Apr 2024 – Oct 2024*
 - Automated ETL with 4 SSIS packages that loaded 7 CSVs into SQL Server via 17 procedures, **cutting manual effort**
 - Delivered 70K-record dataset powering churn model at **97.4% accuracy**, enabling targeted at-risk customer retention

EDUCATION

B.Sc. Computer Science | Faculty of Computers & Artificial Intelligence, Fayoum University *Oct 2022 – Jun 2026*
GPA: 3.42 / 4.0 | **Core Coursework:** Big Data Engineering, Cloud Computing, Database Management Systems, Advanced SQL, Object-Oriented Programming, Data Structures & Algorithms, Machine Learning, Operating Systems (Linux, Docker), Python Programming, Data Visualization (Power BI)

PROJECTS

- **Retail Lakehouse Pipeline [GitHub](#)** | PostgreSQL, Databricks, dbt, Airflow, S3, Docker, Power BI, GitHub Actions
 - Built ELT pipeline with two **Bronze** ingestion paths (PostgreSQL tables, S3 CSVs), transformed via dbt into Silver/Gold layers
 - Modeled **Gold as a Galaxy Schema** with **point-in-time SCD2 resolution**, **caught and fixed a 10x fan-out bug** through mandatory grain verification, and passed 123 automated dbt tests gated by GitHub Actions CI
 - Orchestrated the pipeline in Airflow (on Docker), parallelizing independent branches after the shared date dimension task for a **measured ~16% wall-clock reduction** across 3 runs versus a sequential baseline
 - Delivered a **dual consumption layer**: a three-page Power BI report (Sales, Supplies, Workforce), and a Databricks Genie Agent enabling natural language ad-hoc queries against the gold layer
- **E-Commerce Lakehouse Databricks [GitHub](#)** | Databricks, PySpark, Unity Catalog, Jobs
 - Built an end-to-end **cloud data lakehouse pipeline** processing 200K+ records from 6 data sources through a 3-task Databricks Jobs DAG (**Bronze** ingestion → **Silver** transformation → **Gold** star schema)
 - Delivered a governed, analytics-ready Delta Lake platform using **Medallion Architecture** and **Unity Catalog**, enabling downstream **OLAP consumption** via dimensional **star schema** modeling
- **SQL Data Warehouse Project [GitHub](#)** | SQL Server, T-SQL, Star Schema, Medallion Architecture
 - Architected a fully automated warehouse pipeline ingesting 60K+ records across 6 sources via 2 master stored procedures, applying OLTP→OLAP separation, star schema design, and idempotent ETL execution
- **EduMate RAG [Live](#) | [GitHub](#)** | FastAPI, LangChain, Qdrant Cloud, ChromaDB, Hugging Face Spaces
 - Independently built EduMate-RAG, a decoupled microservice (**platform-integrated + standalone modes**) with **hybrid search** over Qdrant Cloud in production and used ChromaDB for dev, indexing 24K+ document chunks via automated PDF pipeline

CERTIFICATIONS

Huawei HCIA Big Data Associate | Huawei Technologies Co., Ltd *Issued: Mar 2026 | Valid: Mar 2029*
AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services *Issued: May 2026 | Valid: May 2029*

TECHNICAL SKILLS

- **Languages:** SQL, Python
- **Data Engineering & Orchestration:** ETL/ELT, Apache Airflow, dbt, Medallion Architecture, Data Quality
- **Big Data & Distributed Computing:** Databricks, Spark, Lakeflow Connect, Hadoop (HDFS, YARN, Hive), Kafka
- **Data Warehousing & Modeling:** Dimensional Modeling (Star, Snowflake, Galaxy Schema), Unity Catalog, Delta Lake
- **Databases:** SQL Server, PostgreSQL, MySQL, HBase
- **Cloud (AWS):** S3, EC2, RDS, VPC, IAM, Lambda, Glue, Redshift, ELB, Auto Scaling
- **Tools:** Git/GitHub, Docker, CI/CD (GitHub Actions), FastAPI, LangChain, Power BI